IMPACT LAB · RESEARCH DAY

BauCode

a wider lens on material use for sustainability and well-being for engineers, architects and researchers

· 7 Nov 2026 ·

Contents

01 Research question 3

02 Demo 4

03 Team 5

04 Arguments: two lenses 6

05 Golden eval 7

06 Research loop 8

07 Takeaways 9

08 References 10

RESEARCH QUESTION

How can design and engineering teams analyze and compare a material details and combinations by its embodied carbon and by what it does for the people?

Embodied carbon over the life cycle
Six structural systems

Occupant well-being and health
Indoor air, moisture buffering, acoustic comfort

Compatibility of the build-up
Vapour-diffusion order, condensation, junctions

Visible gaps

People spend about 87 % of their time indoors [4]; the WHO links housing quality directly to health [5].

Demo & Research Loops

Live demo: veterdney.github.io/claude-impact-lab-2026

Layer 2 at 100 mm

206

kg CO_2e, whole wall



Layer 2 at 160 mm

260

kg CO_2e, whole wall

Takeaways for your next building

1 **Treat code as a gate**

2 **Read the system boundary**

Check declared unit, modules and validity; keep module D apart [8].

3 **Inspect interior layers**

Clay or timber surfaces buffer humidity; ask for emission tests [3, 6].

4 **Detail the junctions**

Check vapour-diffusion order and condensation plane per build-up [9].

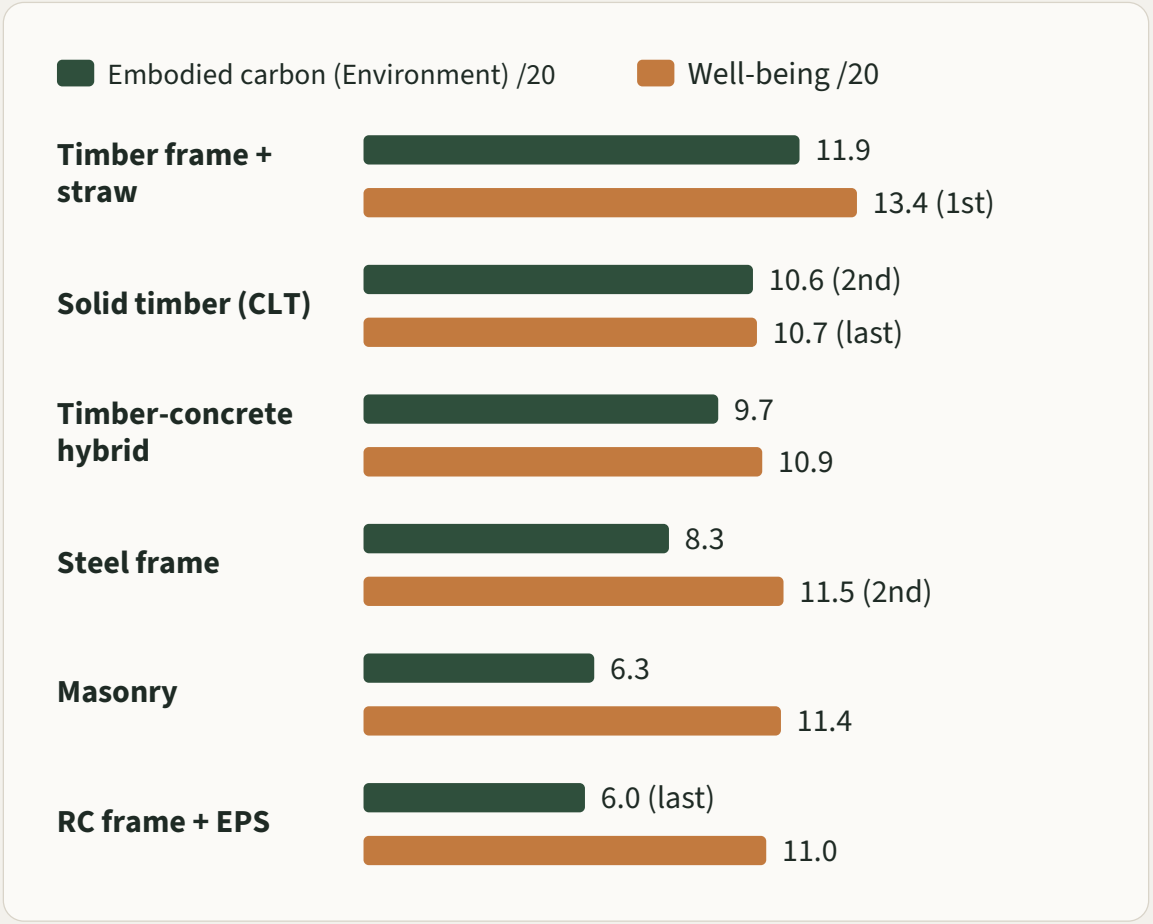
5 **Design for deconstruction**

bonded ETICS is replaced whole at 40 years [11].

6 **Verify bio-based build-ups**

Straw needs hygrothermal proof; keep timber + wood fibre as fallback [10].

Two lenses on six structural systems



The occupant lens reorders the field

Solid timber is 2nd on carbon but last on well-being; the clay-plastered straw wall leads both by buffering indoor humidity [3, 7].

Health data is the thinnest layer

ÖKOBAUDAT carries no VOC, buffering or acoustic values [1]; well-being ratings stay assumed until tested [6].

Golden eval: how we built it

Frozen sources

ÖKOBAUDAT export of 7 Nov 2026 (44,973 rows) and the lab's result files

Computed gold

A script derives every answer from the data; nothing is typed by hand

Gates first

Invented numbers, unit errors or module D in a total fail outright

Graded k times

Code checks, then an LLM judge; reported as pass@1 and pass^k

10	8	10	11	9	8	10	6
Lookup	Unit conversion	Data traps	Missing data	Assembly calculation	Ranking	Design advice	Adversarial

72
cases in 8 families

248
agent runs per full pass

100 %
self-test: 145 gold runs pass, 48 corrupted fail

5
CLT datasets our own analysis missed

Research loop: 8 stages, 5 rounds

1 Frame
Scope, criteria, data rules

2 Profile data
44,973 rows, cleaned

3 Build model
Build-ups, Glaser check

4 Score
Layer › part › building

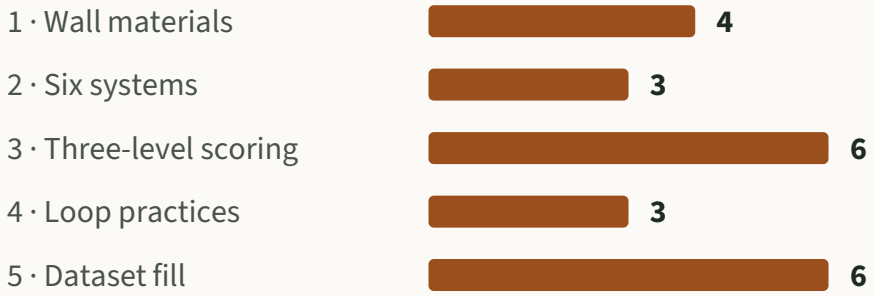
5 Cross-check
Scenarios, 400 draws

6 Verify claims
Every sentence vs a table

7 Report
Files, page, pull request

8 Record
Tasks, journal, commit

22 errors the loop caught itself



State lives in files, not memory

- A checkable goal in PROMPT.md
- Prioritised TASKS.md and dated JOURNAL.md
- Test written first and seen failing
- One subagent per atomic step
- One commit per finished task

Each round’s result becomes the next round’s brief · interactive map: veterdney.github.io/claude-impact-lab-2026/loop.html

Team



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Roles are supplied labels, not claims about authorship of particular changes.

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